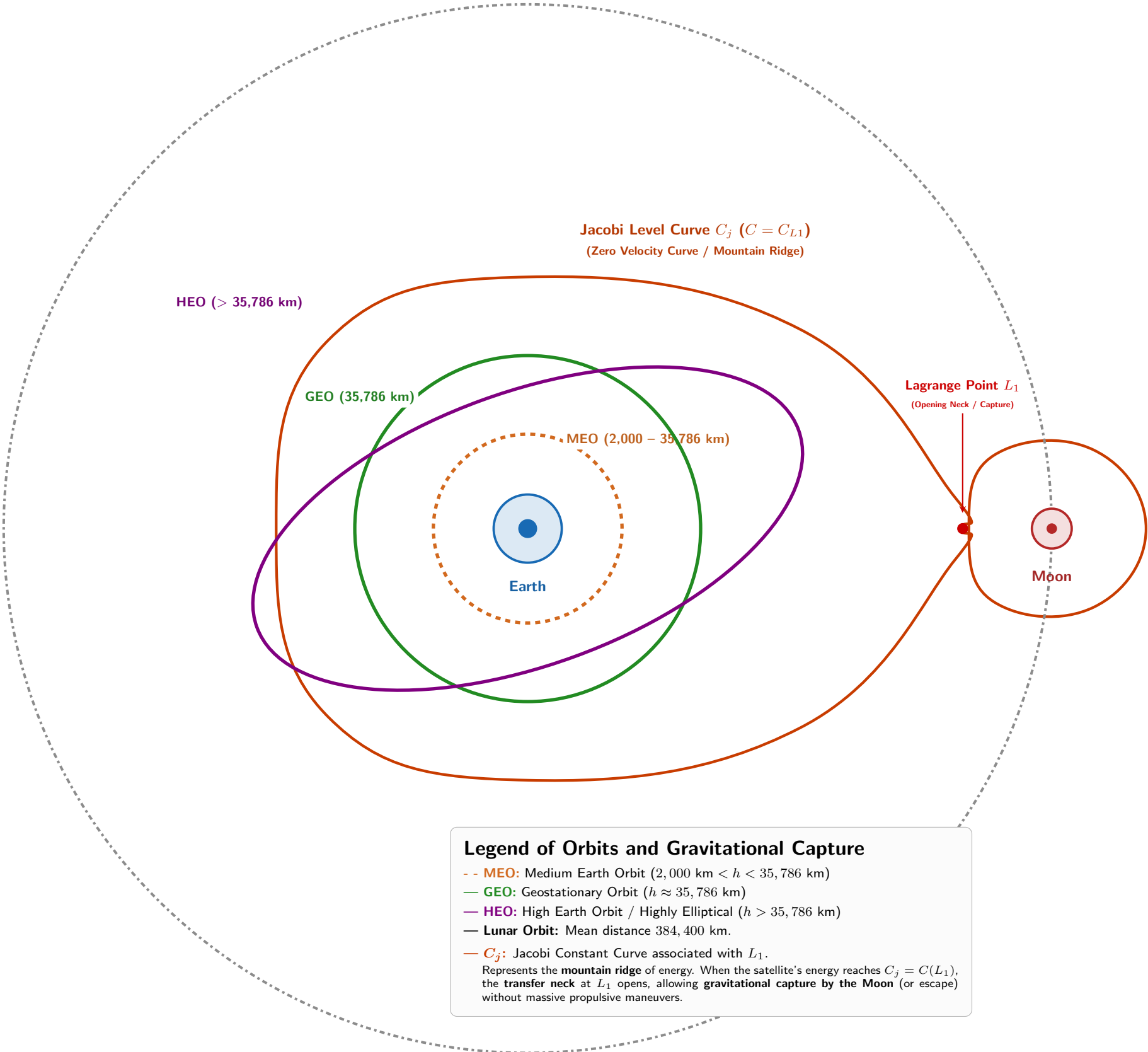




Jacobi Level Curve C_j ($C = C_{L_1}$)
(Zero Velocity Curve / Mountain Ridge)

HEO ($> 35,786$ km)

GEO (35,786 km)

MEO (2,000 – 35,786 km)

Earth

Lagrange Point L_1
(Opening Neck / Capture)

Moon

Legend of Orbits and Gravitational Capture

- - **MEO**: Medium Earth Orbit ($2,000 \text{ km} < h < 35,786 \text{ km}$)
- **GEO**: Geostationary Orbit ($h \approx 35,786 \text{ km}$)
- **HEO**: High Earth Orbit / Highly Elliptical ($h > 35,786 \text{ km}$)
- **Lunar Orbit**: Mean distance 384,400 km.
- **C_j** : Jacobi Constant Curve associated with L_1 .

Represents the **mountain ridge** of energy. When the satellite's energy reaches $C_j = C(L_1)$, the **transfer neck** at L_1 opens, allowing **gravitational capture by the Moon** (or escape) without massive propulsive maneuvers.

Lunar Orbit ($\approx 384,400$ km)